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 Studierichting: Informatica
 Oefeningenreeks 3F nr. 2.4

$$f(x) = \frac{x^3 + 3x^2 - 4x + 2}{x^2 - 4}$$

1. Euclidische deling:

$$\begin{array}{r|l} \begin{array}{rrrr} x^3 & +3x^2 & -4x & +2 \\ - & x^3 & & \\ \hline & 3x^2 & -4x & +2 \\ - & 3x^2 & & -12 \\ \hline & & & 14 \end{array} & \begin{array}{l} x^2 - 4 \\ x + 3 \end{array} \end{array}$$

$$\Rightarrow f(x) = x + 3 + \frac{14}{x^2 - 4}$$

2. SA:

$$gr(A) = gr(B) + 1$$

$$\Rightarrow y = x + 3$$

3. VA:

$$x^2 - 4 = 0$$

$$x = \pm 2$$

$$\left\{ \begin{array}{l} \lim_{x \rightarrow -2^-} f(x) = +\infty \\ \lim_{x \rightarrow -2^+} f(x) = -\infty \end{array} \right.$$

$$\left\{ \begin{array}{l} \lim_{x \rightarrow 2^-} f(x) = -\infty \\ \lim_{x \rightarrow 2^+} f(x) = +\infty \end{array} \right.$$

$$\Rightarrow x = -2 \text{ en } x = 2$$

4. Tekenonderzoek:

x	-2	2
$f(x) - (x + 3)$	+	+
$x \rightarrow -\infty$	-	$x \rightarrow +\infty$
$\Rightarrow f$ boven SA		$\Rightarrow f$ boven SA

References